

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2014

EE216 Integrated Circuits & Analysis

Date: 10.05.2014, Saturday

Time: 10:00a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 What do you mean by IC? Give the classification of ICs according to their mode of operation and according to the number of components. [07]

Q - 2 (a) Draw and explain the equivalent circuit of an op-amp. Also draw and explain the ideal voltage transfer curve. [04]

(b) Find out the closed-loop voltage gain A_F for noninverting Amplifier with negative feedback with neat circuit diagram. [06]

(c) Draw and explain voltage to current converter with floating load. [04]

OR

Q - 2 (a) Design a differentiator to differentiate an input signal that varies in frequency from 10 Hz to about 2 kHz. If a sine wave of 1 V peak at 2000 Hz is applied to that circuit draw its output and input waveforms. (Take $C_1 = 0.1 \mu\text{F}$) [08]

(b) For the noninverting Amplifier if $R_F = 20 \text{ k}\Omega$ and $R_1 = 1 \text{ k}\Omega$, then determine the value of output voltage if the input is a 5 V peak sine wave at 1 kHz. Also sketch the output and input waveform. Assume that V_{OOT} is zero. [04]

(c) What is the meaning of an electric filter? Just write down its classification. [02]

Q - 3(a) Draw and explain Integrator circuit with necessary input and output waveforms. [06]

(b) Draw the frequency response curve of the major active filters. [05]

(c) Design a low-pass filter circuit at a cutoff frequency of 500 Hz with pass band gain of 2. (Take $C = 0.01 \mu\text{F}$). Also draw neat circuit diagram. [03]

OR

Q - 3 (a) Design the square wave oscillator using op-amp so that $f_0 = 500 \text{ Hz}$. The op-amp is a 741 with a DC supply voltages $= \pm 15 \text{ V}$. (Take $C = 0.05 \mu\text{F}$). Also draw neat circuit diagram. [06]

(b) Draw and explain Triangular wave generator circuit using op-amp. [08]

SECTION – II

Q - 4 Write down the electrical characteristics of an ideal op-amp. [07]

Q - 5 (a) Define the following terms with necessary diagrams. [06]

- (i) Input offset Voltage
- (ii) Input bias current
- (iii) Common-Mode Rejection Ratio (CMRR)
- (iv) Supply Voltage Rejection Ratio (SVRR)

(b) Design the circuit such that output of that circuit is Inverting and Average of four input voltages. [04]

(c) Draw and explain phase shift Oscillator. [04]

OR

Q - 5 (a) Draw pin diagram of 555 Timer IC and write down function of each pin. [06]

(b) Using the 7805C voltage regulator, design a current source that will deliver a 0.20 A current to the 40 Ω , 10 W load. Also draw neat circuit diagram. [05]

(c) Draw and explain All-pass filter circuit. [03]

Q - 6 (a) Draw the different circuit arrangements for obtaining positive and negative supply voltages for an op-amp. [06]

(b) Draw and explain Differential Amplifier with one op-amp and also find out its voltage gain. [06]

(c) For the inverting Amplifier if $R_F = 10 \text{ k}\Omega$ and $R_1 = 1 \text{ k}\Omega$, then sketch the output and input waveform if the input is a 1 V peak-to-peak sine wave at 1 kHz. Assume that V_{OOT} is zero. [02]

OR

Q - 6 (a) Draw and explain 555 timer as an astable multivibrator. Also draw necessary waveforms. [08]

(b) Derive the slew rate equation with necessary circuit diagram. [06]
